

Grading Session 16 Final Project

1. Data Wrangling

- Data loaded correctly with Pandas
- Data cleaned, dropped unnecessary columns
- Renamed columns for clarity
- Checked for and found no missing values
- Thorough inspection, inspected data types

10/10 efficient, correct

2. Exploratory Data Analysis

- Discussed effects of sorted draws for fairness of assessment
- Visualizations:
 - i. Bar chart indicating how each number is statistically fair
 - ii. Histogram of sum of numbers showing normal looking distribution
 - iii. Line chart showing lack of seasonality or time trend
 - iv. Box plot highlighting the effect of sorting, middle position naturally has the most variation
 - v. Bar chart showing frequency of last digits distributed roughly uniform, with natural variation
 - vi. Word Cloud should be fairly even, but is showing a few numbers as the largest due to randomness (noise)

14/15 clear insights and visuals, a comparative frequency plot for all numbers could have added to the insights

3. Feature Engineering

- Created aggregated frequency features
- Data splitting done for training, validation, and testing
- Data scaling with Standard Scaler

10/10 well prepared and documented

4. Modeling

- Implemented multiple models: Logistic Regression, Random Forest, Decision Tree, Linear Regression, KMeans Clustering
- Logical choices for model and parameters

15/15 well done, very clean

5. Model Evaluation

- Used accuracy, F1, confusion matrix, ROC AUC

10/10 thorough insight, could discuss overfitting

6. Model Analysis

- Multiple models built and interpreted
- Feature importance chart shows how each feature contributes
- Noted that random nature of lottery limits prediction accuracy
- Compared side by side evaluation of different models

10/10 in depth analysis of each model's results

7. Code runs without Error

10/10 no errors

8. Readability and Documentation

10/10 well documented, good narrative flow (Intro -> EDA -> Modeling -> Conclusion)

9. Final Conclusion

- Summarized key findings
- Linked conclusions to objectives

10/10 conclusive and insightful

99/100 very well done, thorough and in depth analysis, logical application of tools for data modeling